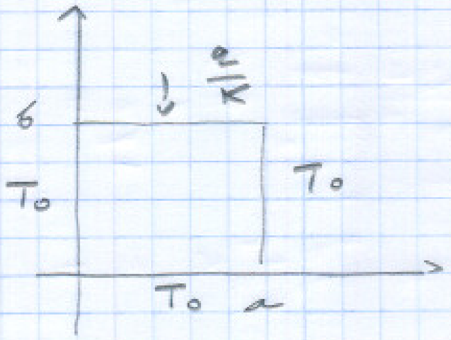
מועד א' תשס"ו

1.

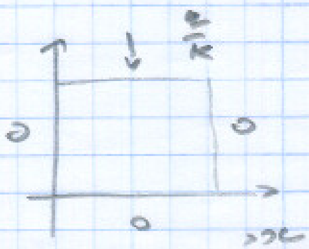


$$\Delta T = 0 \qquad \left.\frac{\partial T}{\partial y}\right|_{y=b} = \frac{q}{K}$$

נגדיר $\bar{J} = -K\bar{\nabla}T$

נחפש פתרון פרטי. $T_p(x) = T_0$

ונפתור את הבעיה:



לא מקיימים תנאי שפה

פתרון כ״ל $\cancel{e^{-\lambda y}\cos\lambda x} \quad \cancel{e^{\lambda y}\cos\lambda x}$

$$e^{-\lambda y}\sin\lambda x \quad e^{\lambda y}\sin\lambda x$$

$$\Rightarrow \quad T_\lambda(x,y) = A_\lambda e^{-\lambda y}\sin(\lambda x) + B_\lambda e^{\lambda y}\sin(\lambda x)$$

קבלנו $0 = T(x=a,y) \Rightarrow \sin(\lambda a) = 0$

$$\lambda a = \pi n$$

$$\lambda = \frac{\pi n}{a} \qquad n = 1,2,3,\ldots$$

$$0 = T_\lambda(x, y=0) \Rightarrow (A_\lambda + B_\lambda) = 0$$

$$\Rightarrow \quad T(x,y) = \sum_{n=1}^{\infty} A_n \sinh\left(\frac{\pi n y}{a}\right)\sin\left(\frac{\pi n y}{a}\right)$$

[illegible] לפי [illegible]:

$$\frac{q}{K} = \frac{\partial T}{\partial y}\Big|_{y=b} = \sum_{n=1}^{\infty} A_n \frac{\pi n}{a}\cosh\left(\frac{\pi n}{a} b\right)\sin\left(\frac{n\pi x}{a}\right)$$

$$\Rightarrow \sum_{n=1}^{\infty} A_n \sin\left(\frac{n\pi x}{a}\right) = \frac{q}{K}\,\frac{a}{\pi n}\,\frac{1}{\cosh\left(\frac{\pi n b}{a}\right)} \equiv D_n$$

$$\Rightarrow A_n = \frac{2}{a}\int_0^a D_n \sin\left(\frac{n\pi x}{a}\right)dx$$

$$= \frac{2D_n}{a}\left[\frac{-\cos\left(\frac{n\pi x}{a}\right)}{\left(\frac{n\pi}{a}\right)}\Bigg|_0^a\right]$$

$$= \frac{2D_n}{\left(\frac{n\pi}{a}\right)a}\Big[\underbrace{-\cos(\pi n)}_{(-1)^{n+1}}+1\Big] = \frac{4D_n}{\pi n}$$

$n = 2K+1$

$$\underset{n=2K+1}{A_n} = \frac{4}{\pi n}\,\frac{qa}{K\pi n}\,\frac{1}{\cosh\left(\frac{\pi n b}{a}\right)}$$

הפתרון הסופי:

$$= \frac{4qa}{\pi^2 K n}\,\frac{1}{\cosh\left(\frac{\pi n b}{a}\right)}$$

$$\Rightarrow T(x,y) = T_0 + \sum_{\substack{\text{odd } n \\ n=1,3,5}} \frac{4qa}{\pi^2 K n}\,\frac{\sinh\left(\frac{\pi n y}{a}\right)}{\cosh\left(\frac{\pi n b}{a}\right)}\sin\left(\frac{n\pi x}{a}\right)$$

$$\frac{\partial T}{\partial t}=\kappa\frac{\partial^2 T}{\partial x^2}$$ .2

$T(x,0)=A\delta(x-\xi)$ $\qquad -\infty<x<\infty$

$$G(x,\zeta,t)=\frac{1}{\sqrt{4\pi\kappa t}}e^{-\frac{(x-\zeta)^2}{4\kappa t}}$$

$$\Rightarrow T(x,t)=\int_{-\infty}^{\infty}A\delta(\zeta-\xi)\frac{1}{\sqrt{4\pi\kappa t}}e^{-\frac{(x-\zeta)^2}{4\kappa t}}d\zeta$$

$$=A\frac{1}{\sqrt{4\pi\kappa t}}e^{-\frac{(x-\xi)^2}{4\kappa t}}$$

הזמן בו בנק' $x=x_0$ מקבלת מקסימום:

$$T(x=x_0)=\frac{A}{\sqrt{4\pi\kappa t}}e^{-\frac{(x_0-\xi)^2}{4\kappa t}}$$

$$0=\frac{\partial T}{\partial t}\Big|_{x=0}=-\frac{1}{2}(4\pi\kappa t)^{-\frac{3}{2}}\cancel{A}\cancel{e^{-\frac{(x_0-\xi)^2}{4\kappa t}}}$$

$$+\frac{\cancel{A}}{\sqrt{4\pi\kappa t}}\cancel{e^{-\frac{(x_0-\xi)^2}{4\kappa t}}}\cdot\left[-\frac{(x_0-\xi)^2}{4\kappa}\cdot\left(-t^{-2}\right)\right]$$

$$\Rightarrow \frac{1}{2}(4\pi\kappa t)^{-\frac{3}{2}}=\frac{(x_0-\xi)^2}{4\kappa t^2}\cdot\frac{1}{\sqrt{4\pi\kappa t}}$$

$$\frac{1}{(4\pi\kappa)t}\quad \frac{1}{2}(4\pi\kappa t)^{-1}=\left(\frac{(x_0-\xi)^2}{2\kappa t^2}\right)$$

$$\Rightarrow t=t_{max}=\frac{(x_0-\xi)^2}{2\kappa}$$

$$T_{max}=T(x_0,t_{max})=\frac{A}{\sqrt{4\pi\kappa\cdot\frac{(x_0-\xi)^2}{2\kappa}}}e^{-\frac{1}{2}}$$

$$=\frac{A}{\sqrt{\frac{\pi}{2}(x_0-\zeta)^2}}e^{-\frac{1}{2}}$$

$x=0$ ———————— $x=L$ .3

$u_{tt}=c^2u_{xx}$

$$u(x,0)=0 \qquad u(x=0,t)=u(x=L,t)=0$$

$$\left.\frac{\partial u}{\partial t}\right|_{t=0}=\frac{P}{\rho}\delta(x-x_0)$$

פתרון כללי:

$$u(x,t)=\sum_{n=1}^{\infty}(A_n\sin(\frac{n\pi x}{L})\sin(\frac{n\pi ct}{L})$$

$$\left.\frac{\partial u}{\partial t}\right|_{t=0}=\sum_{n=1}^{\infty}A_n\sin(\frac{n\pi x}{L})\frac{n\pi c}{L}\underset{1,\ t=0}{\underset{\shortparallel}{\cos(\frac{n\pi ct}{L})}}=\frac{P}{\rho}\delta(x-x_0$$

$$\Rightarrow \quad A_n=\frac{2}{L}\cdot\frac{PL}{\rho n\pi c}\int_0^L\delta(x-x_0)\sin(\frac{n\pi x}{L})\,dx$$

$$=\frac{2P}{\rho\pi nc}\sin(\frac{n\pi x_0}{L})$$

$$u(x,t)=\sum_{n=1}^{\infty}\frac{2P}{\rho\pi nc}\sin(\frac{n\pi x_0}{L})\sin(\frac{n\pi x}{L})\sin(\frac{n\pi ct}{L})$$

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$$u(x)=\cos(x)+\lambda\int_0^{\pi}\sin(x)\,u(t)\,dt$$

$$=\cos(x)+\lambda\sin(x)I$$

$$\Rightarrow I=\int_0^{\pi}(\cos(t)+\lambda\sin(t)I)\,dt$$

$$=-\lambda I\cos(t)\Big|_0^{\pi}=-\lambda I(-1-1)$$

$$I=2\lambda I$$

$$\lambda=\frac{1}{2}$$

$u(x)=\cos(x)$ $\lambda\neq\frac{1}{2}$ אר

$u(x)=\cos(x)+\frac{1}{2}\cdot A\sin(x)$ $\lambda=\frac{1}{2}$ אר

$\lambda\neq\frac{1}{2}$ בדיקה

$$u(x)=\cos(x)+\lambda\sin(x)\int_0^{\pi}\cos(t)\,dt$$

$$=\cos(x)$$

$\lambda=\frac{1}{2}$ בדיקה

$$u(x)=\cos(x)+\frac{1}{2}\sin(x)\int_0^{\pi}\cos(t)+\frac{1}{2}A\sin(t)\ dt$$

$$=\cos(x)+\frac{\sin(x)}{2}\cdot\frac{A}{2}\left[-\cos(t)\right]\Big|_0^{\pi}$$

$$=\cos(x)+\frac{1}{2}A\sin(x)$$

$$I(y(x))=\int_0^{\pi/2}\left[y'^2-y^2\right]dx$$ 5.

$$F(y,y')=y'^2-y^2$$

משוואת EL:

$$\frac{dF}{dy'}=2y' \quad \frac{d}{dx}\left(\frac{dF}{dy'}\right)=2y'' \qquad \frac{dF}{dy}=-2y$$

$$\Rightarrow\ 2y''-[-2y]=0$$

$$y''+y=0$$

$$y(x)=A\cos(\cancel{k}x)+B\sin(\cancel{k}x)$$

$y(0)=0\ \Rightarrow\ A=0$ תנאי שפה

$$y\left(\frac{\pi}{2}\right)=1\ \Rightarrow\ B\sin\left(\cancel{k}\frac{\pi}{2}\right)=1$$

$$\Rightarrow\ \cancel{k\neq 0}\ \ B=1$$

$$y=\sin(x)$$